

TE AI&DS

Cyber Security (317530)

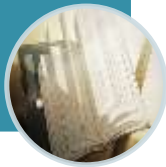
Prepared By:

Mrs. Sonali B.Gavali
Assistant Professor
AI&DS Department
DIT,Pimpri.

Examination Scheme

- 04hrs/week

Lecture



- 03 credits

Credit

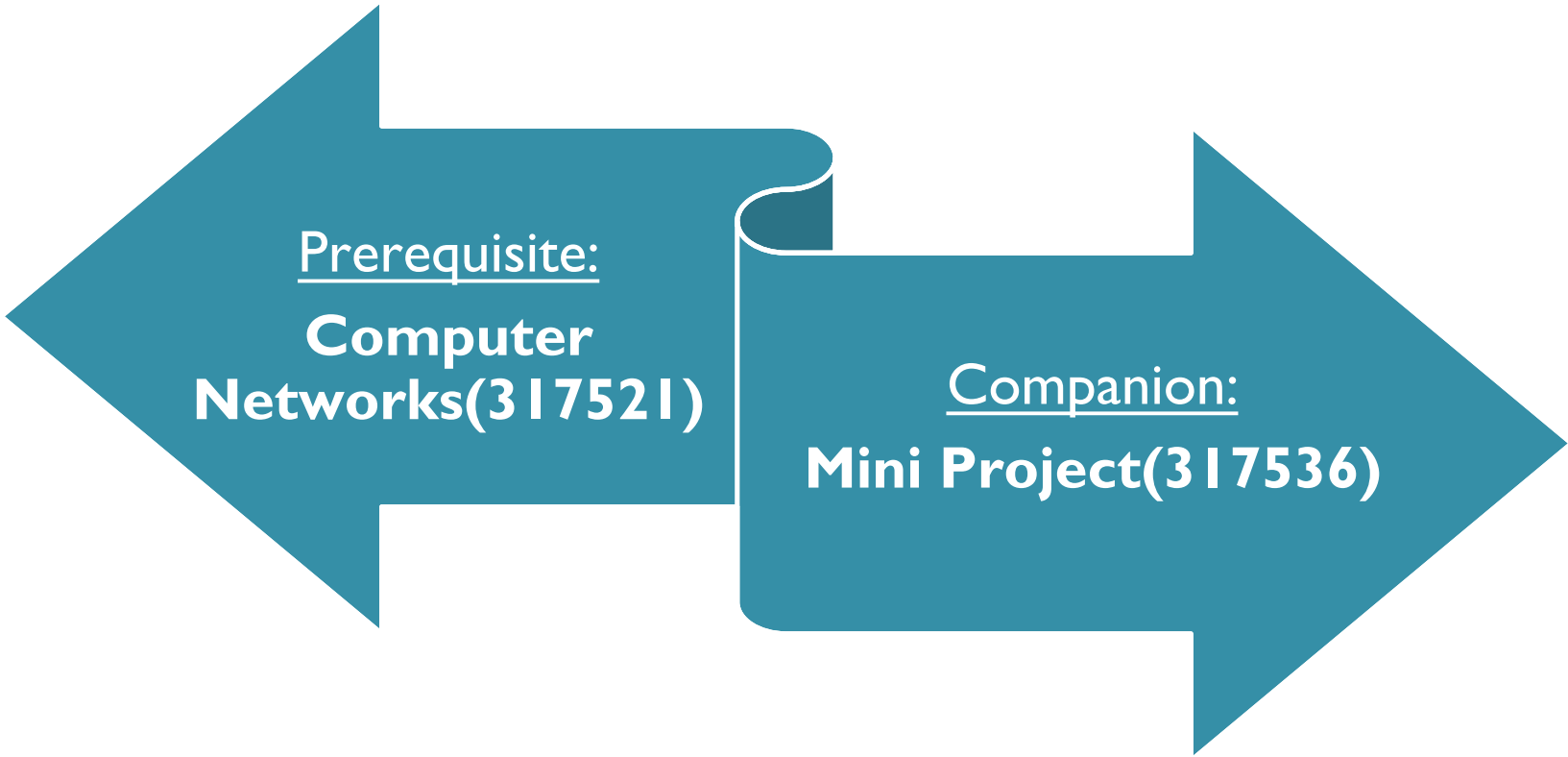


- Insem:30M
- Endsem:70M

Exam



Prerequisite & Companion Courses



The diagram consists of two large, teal-colored arrows pointing in opposite directions. The left arrow points left and contains the text 'Prerequisite: Computer Networks(317521)'. The right arrow points right and contains the text 'Companion: Mini Project(317536)'. The two arrows are connected at their inner ends by a small, white, curved line that resembles a staple or a piece of tape.

Prerequisite:
**Computer
Networks(317521)**

Companion:
Mini Project(317536)

Course Objectives

- To offer an understanding of principle concepts, central topics and **basic approaches in information and cyber security**.
- To know the basics of **cryptography**.
- To acquire knowledge of **standard algorithms and protocols** employed to provide confidentiality, integrity and authenticity.
- To enhance awareness about Personally Identifiable Information (PII), Information Management, cyber forensics.

Course Outcomes

CO1: Gauge the security protections and limitations provided by **today's technology**.

CO2: Identify cyber security **threats**.

CO3: Analyze threats in order **to protect or defend it** in cyberspace from cyber-attacks.

CO4: Build appropriate security solutions against cyber-attacks.



CYBER SECURITY

The protection of
devices, services
and networks –
and the information
on them – from theft
or damage.





Course Content:

Unit-I Introduction

- Introduction,
- Elements of Information Security,
- Security Policy, Techniques, Steps, Categories,
- Operational Model of Network Security,
- Basic Terminologies in Network Security.
- Threats and Vulnerability,
- Difference between Security and Privacy.



I. Introduction

What is Information System



Technology

Agriculture

Information



What is Information System



Information Sources



What is Information System



Information system

- Information system is a combination of software, hardware, and telecommunication networks to collect useful data, especially in an organization.
- Many businesses use information technology to complete and manage their operations, interact with their consumers, and stay ahead of their competition.
- Example....eBay, Amazon, Alibaba, and Google.

Components of Information System

There are 6 components of Information System

- 1) Hardware : - Desktops – Laptops – Mobile
- 2) Software : - Operating systems - Application programs
- 3) Data : - (Information) Facts and figure entered into computers
- 4) Procedures : - How the other components are used
- 5) People: - Users – Technologists - IS support.
- 6) Networks: LAN – WAN – Internet

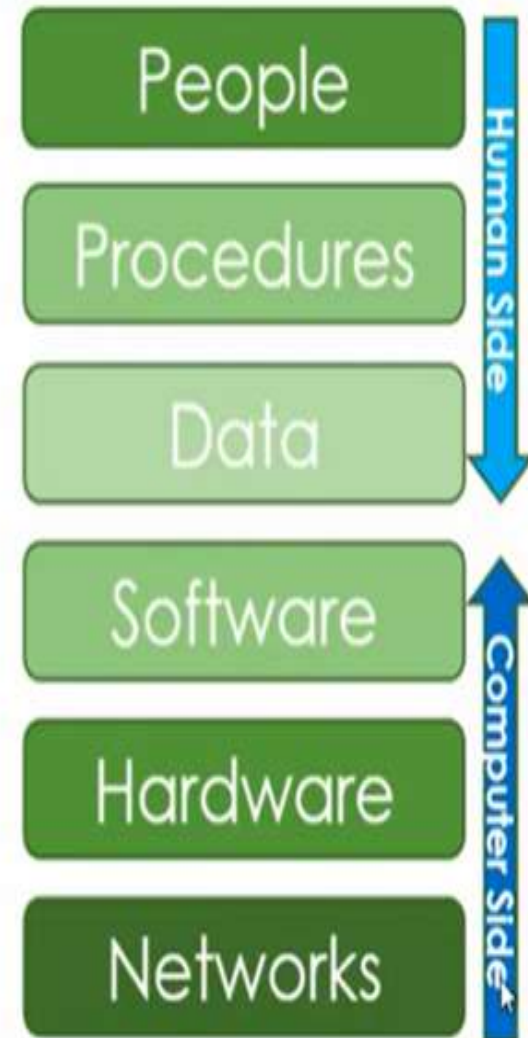
Each component of the Information System also has its own security requirements.



Security requirements of Information System

Each component of the IS also has its own security requirements.

- 1) **Hardware** :- Restricts access to interaction with the hardware, Data has more worth than Laptop
- 2) **Software** : Vulnerabilities in software, bugs
- 3) **Data** :Data stored, processed, and transmitted must be protected
- 4) **Procedures** :Security policies and procedures
- 5) **People**: Weakest link in the security and always threat to Information security
- 6) **Networks**: new security challenges rapidly emerge with networks





2. Elements of Information Security

What is Information Security

- Information Security focuses on **protection of information** including **confidentiality, integrity, and availability**, and includes a variety of protection mechanisms such as **policy, training and awareness programs, and technology**.



CIA Triad



Committee on National Security Systems (CNSS) security model



Elements of Information Security

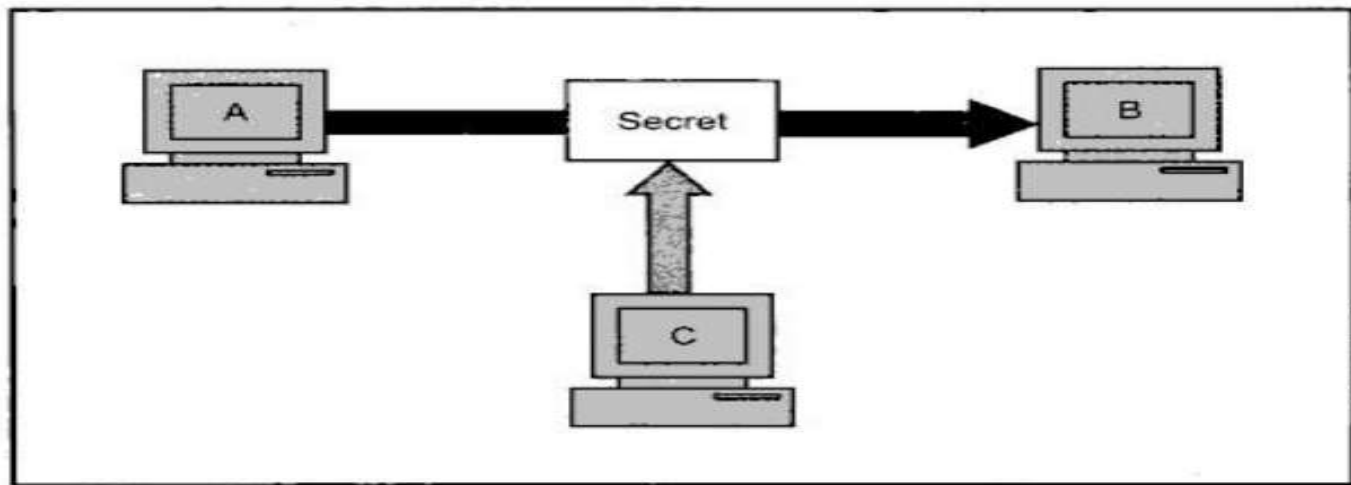
- **Confidentiality**
- **Integrity**
- **Availability**

Elements of Information Security

- **Confidentiality:**

- Means that the information is private and it's not access by any **unauthorized user.(ex. phonepe)**
- Principle –
- Keep personal information private.
- to ensure that it is visible and accessible only to those individuals **who own it or need it to perform their organizational functions.**

- **interception** is where an unauthorized individual gains access to confidential or private information

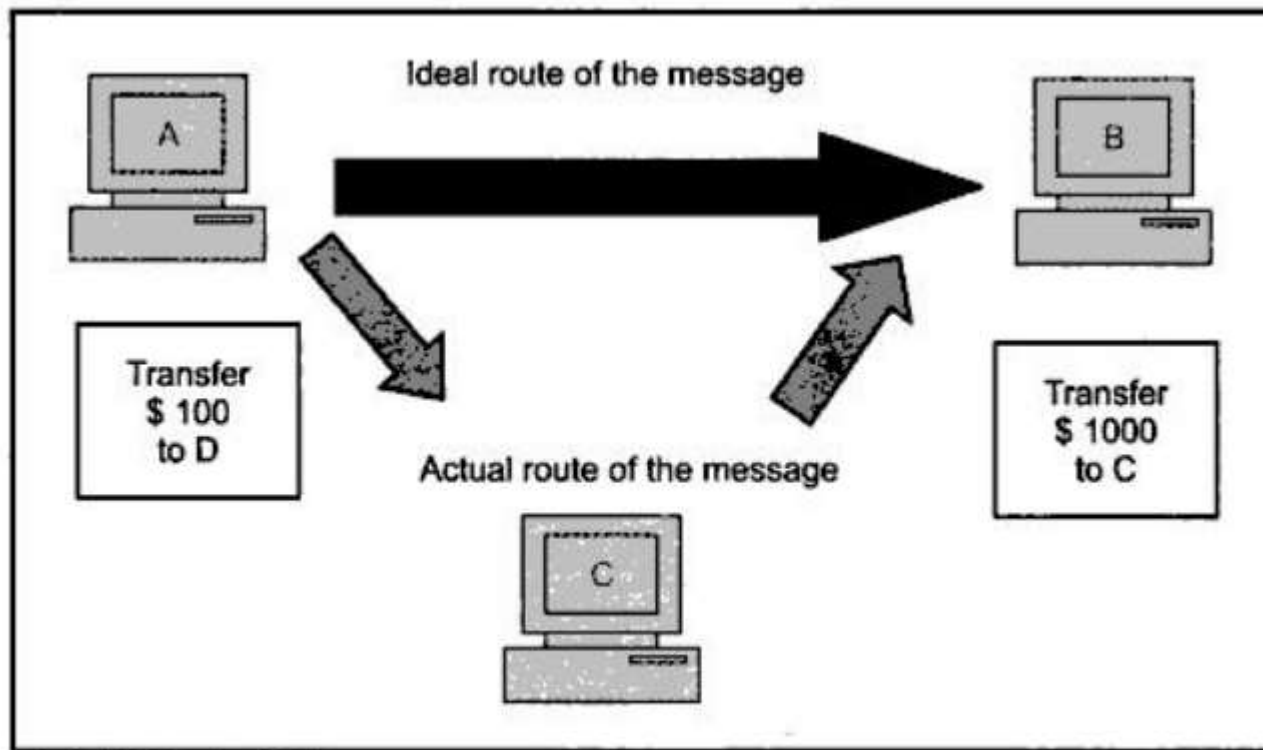


└ Fig. 1.4 *Loss of confidentiality*

- **Integrity:**

- Means that the data/information is not altered/Changed by any unauthorized user.(ex. Money transfer) before it reach to the receiver.
- The principle of integrity ensures that data is accurate and reliable.
- It is not modified incorrectly, whether accidentally or maliciously.

- Modification causes loss of integrity.

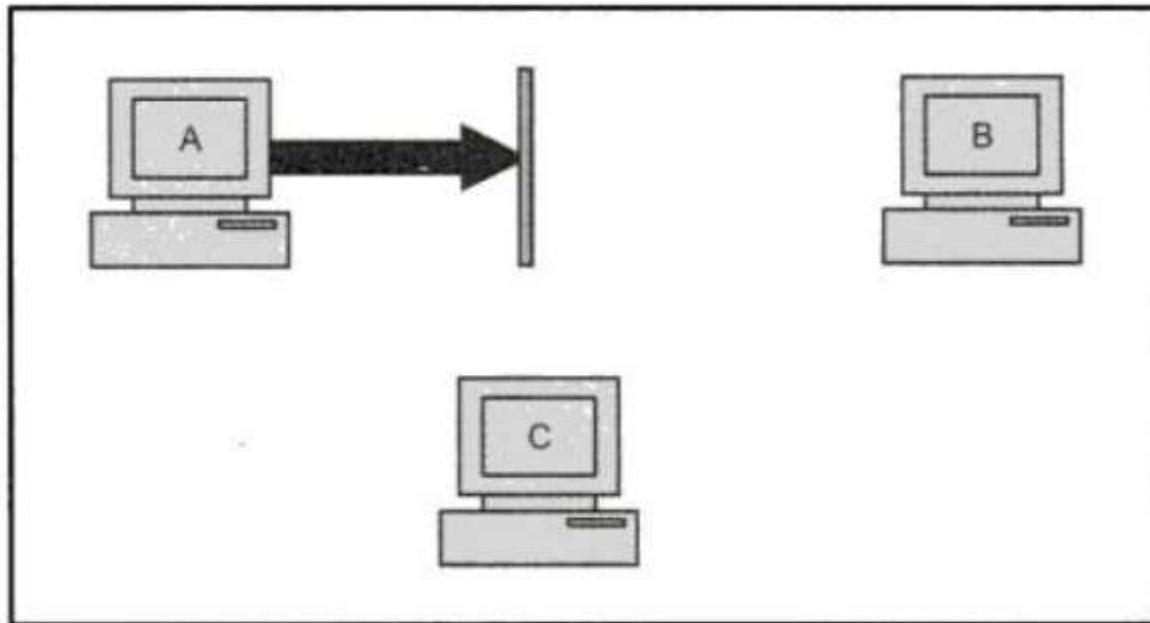


└ Fig. 1.6 *Loss of integrity*

- **Availability:**

- we are able to access the information, when we need it. and there is no disruption of any type of service at the time of accessing the information.(Ex. College Lab)
- The purpose of **availability** is to make the **technology infrastructure**, the applications and the data available when they are needed for an organizational process or for an organization's customers.
- Ex. Denial of service attack(DOS).

- Interruption puts availability of resources in danger.



└ Fig. 1.8 *Attack on availability*

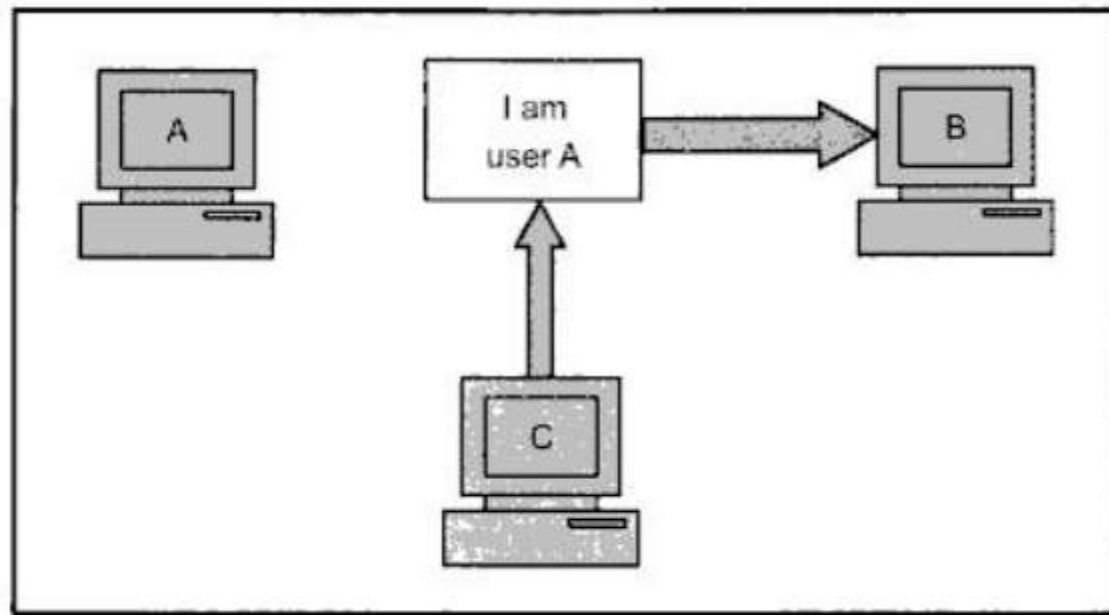
Additional elements....

- **Authentication :**

- This refers to the characteristics of any type of information (like document, communication etc.) is being **genuine** and there is no alteration in it. (college Id)....OR

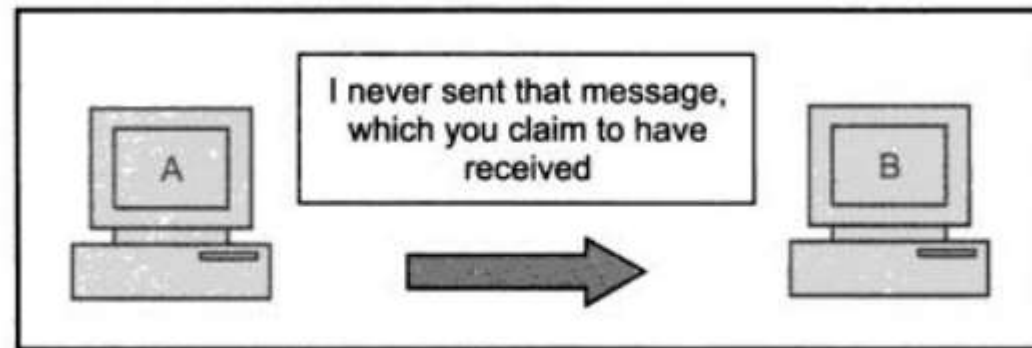
Mechanism helps to establish proof of identities.

- Fabrication is possible in absence of proper Authentication mechanism.



└ Fig. 1.5 *Absence of authentication*

- **Non -Repudiation** : This means that the sender/Receiver of information can not later deny that they do not send/Receive the information.(Ex. Exam form)



└ Fig. 1.7 Establishing non-repudiation

- 
- Access control
 - Specifies who can access what?
 - Works on two areas: rule and role management.
 - Role management: user side: user can do what?
 - Rule management: resource side: resource is accessible and under what circumstances.



Security Policy

Security Policy

- Security policies are a formal set of rules to access company technology and information assets comply with rules and guidelines related to the security of information.
- It is a written document in the organization which is responsible for how to protect the organizations from threats and how to handles them when they will occur.
- A security policy also considered to be a "living document" which means that the document is never finished, but it is continuously updated as requirements of the technology and employee changes.

Need of Security policies-

1) It increases efficiency-

- To increase the level of **consistency** which saves **time, money and resources**.
- The policy should inform the **employees about their individual duties, and telling them what they can do** and what they cannot do with the organization sensitive information.

2) It upholds discipline and accountability-

- When any **human mistake** will occur, will back up any disciplinary action and also supporting a case in a **court of law**.
- The organization policies act as a **contract** which proves that an organization has taken steps to protect its intellectual property, as well as its **customers and clients**.

3) It can make or break a business deal-

- copy of their information security policy to other vendors during a business deal that involves the transference of their sensitive information.
- It is true in a case of bigger businesses which ensures their own security interests are protected when dealing with smaller businesses which have less high-end security systems in place.

4) It helps to **educate employees** on security literacy

- A well-written security policy informs the readers about their importance of **responsibility in protecting the organization sensitive data**. Ex. ERP, college website.
- It involves on choosing the right passwords, to **providing guidelines for file transfers** and data storage which increases employee's overall awareness of security and how it can be strengthened.

Actual Security Policy-VFILEAEH

1. Virus and Spyware Protection policy

- It helps to detect, removes, and repairs the side effects of viruses and security risks by using signatures.
- It helps to detect the threats in the files which the users try to download by using reputation data from Download Insight.
- It helps to detect the applications that exhibit suspicious behavior.



2. Firewall Policy

- It blocks the **unauthorized users** from accessing the systems and networks that connect to the Internet.
- It detects the attacks by cyber criminals.
- It **removes the unwanted sources** of network traffic.



3. Intrusion Prevention policy

This policy automatically detects and blocks the network attacks and browser attacks.

It also protects applications from vulnerabilities.

It checks the contents of one or more data packages and detects malware which is coming through legal ways.



4. LiveUpdate policy

This policy can be categorized into two types one is LiveUpdate Content policy, and another is LiveUpdate Setting Policy.

The LiveUpdate policy contains the setting which determines when and how client computers download the content updates from LiveUpdate.

We can define the computer that clients contact to check for updates and schedule when and how often clients computer check for updates.

5. Application and Device Control

- To Protect from peripheral devices that can attach to a system. ex.PD
- The device control policy applies to both Windows and Mac computers whereas application control policy can be applied only to Windows clients.



6. Exceptions policy

This policy provides the ability to exclude applications and processes from detection by the virus and spyware scans.

7. Host Integrity policy

to keep enterprise networks and data secure.

We use this policy to ensure that the client's computers who access our network are protected and compliant with companies' securities policies.

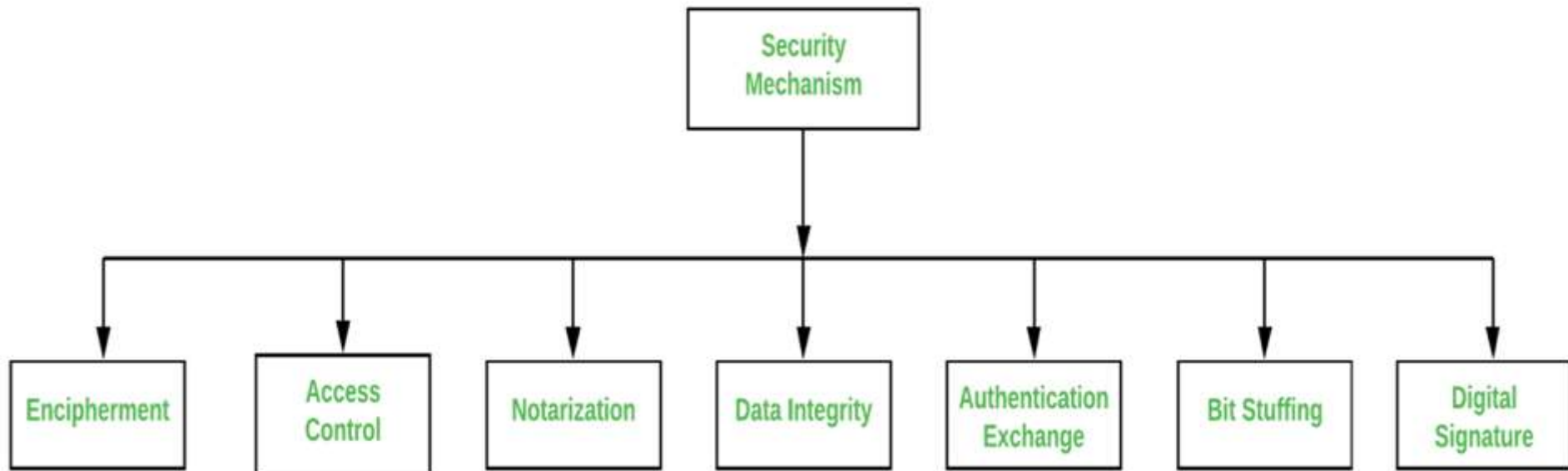
This policy requires that the client system must have installed antivirus.



Cyber Security Techniques

Types of Security Mechanism

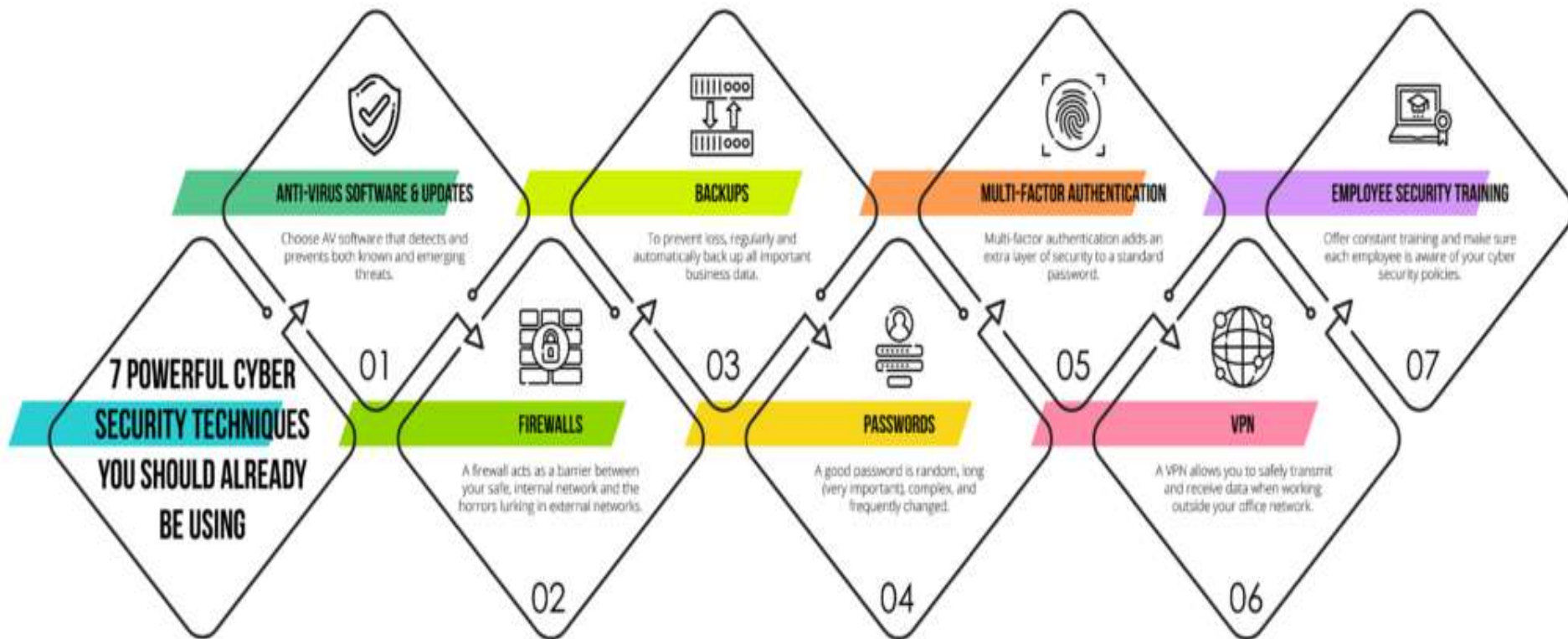
- Network Security is field in computer technology that deals with security of **computer network infrastructure**.
- As the network is necessary for sharing of information at hardware level, or at software level. Therefore security mechanism can termed as is **set of processes that deal with recovery from security attack**.



7 Powerful cyber security techniques

- **Anti-Virus Software & Updates**
- **Firewalls**
- **Backups**
- **Passwords**

- **Multi-Factor Authentication**
- **virtual private network (VPN)**
- **Employee Security Training**



Types of Security Mechanism are :

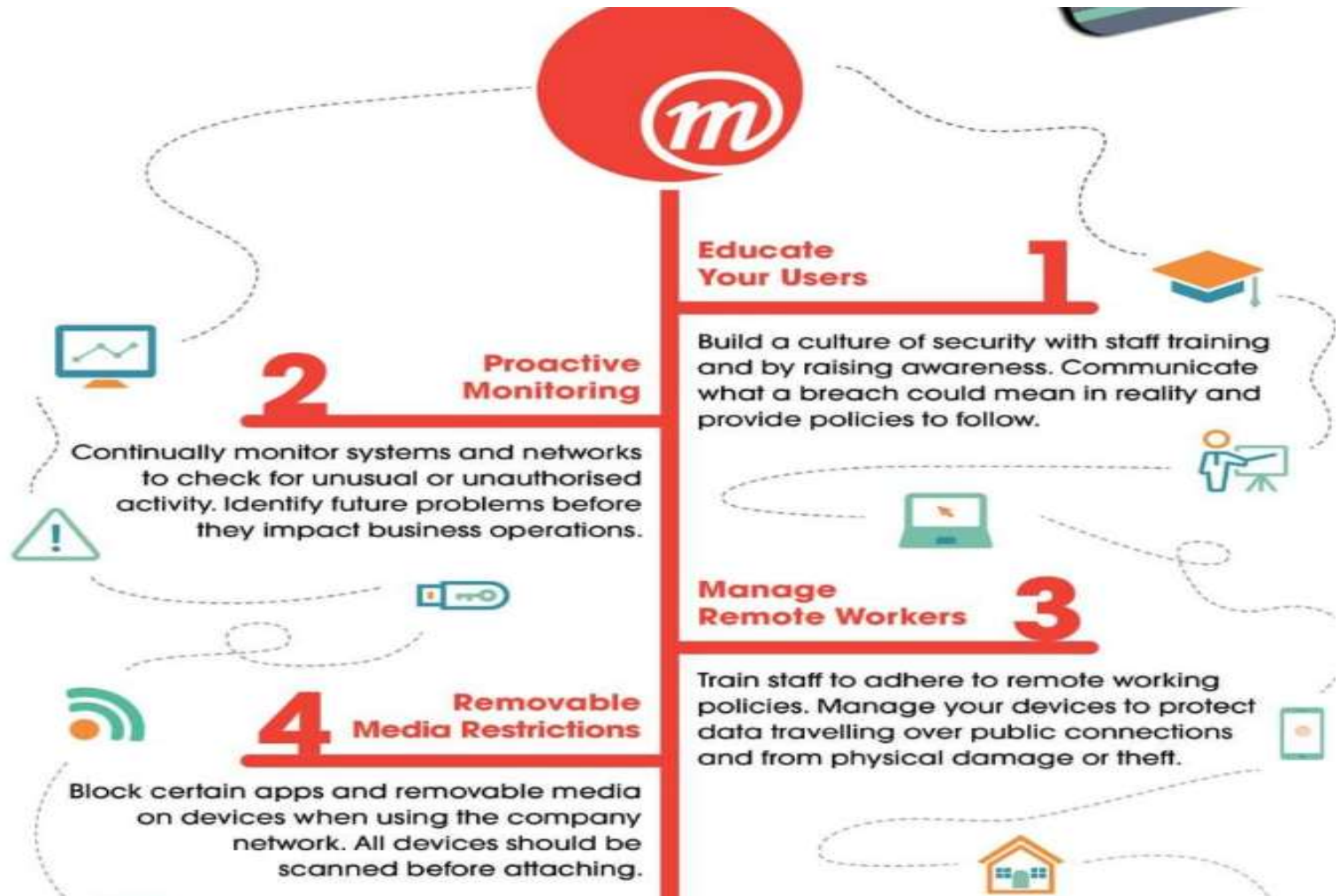
- **Encipherment** :Hiding and covering of data which helps data to become confidential.
- It is achieved by applying algorithms which reconstruct information into not readable form. It is achieved by techniques named Cryptography and Encipherment.
- Level of data encryption is dependent on the algorithm used for encipherment.
- **Access Control** :To stop unattended access to data which you are sending.
- It can be achieved by various techniques such as applying passwords, using firewall, or just by adding PIN to data.

Notarization

- Use of trusted **third party** in communication.
 - It acts as **mediator between sender and receiver** so that if any chance of conflict is reduced.
 - This mediator keeps record of requests made by sender to receiver for later denied.
-
- **Data Integrity :**
 - Appending value to data to which is **created by data itself**.
 - It is similar to **sending packet of information** known to both **sending and receiving parties** and checked before and after data is received.
 - When this packet or data which is appended is checked and is the same while sending and receiving **data integrity is maintained**.

- **Authentication exchange** : Identity to be known in communication.
- This is achieved at the TCP/IP layer where two-way handshaking mechanism is used to ensure data is sent or not.
- **Bit stuffing** : This security mechanism is used to add some extra bits into data which is being transmitted.
- It helps data to be checked at the receiving end and is achieved by Even parity or Odd Parity.
- **Digital Signature** :
 - Adding digital data that is not visible to eyes.
 - It is form of electronic signature which is added by sender which is checked by receiver electronically.

Information security steps





6

Strong User Privileges

Control access to your organisation's data through user account privileges. Only give administrative access where job function dictates a need for this.



8

Malware and Virus Prevention

Install and maintain anti-malware defences. Educate employees in how to be vigilant against possible threats. Keep all software up to date.



Patch Management

5

Apply security patches once released and fully tested for compatibility with your system. Keep an inventory of all devices and their configurations.



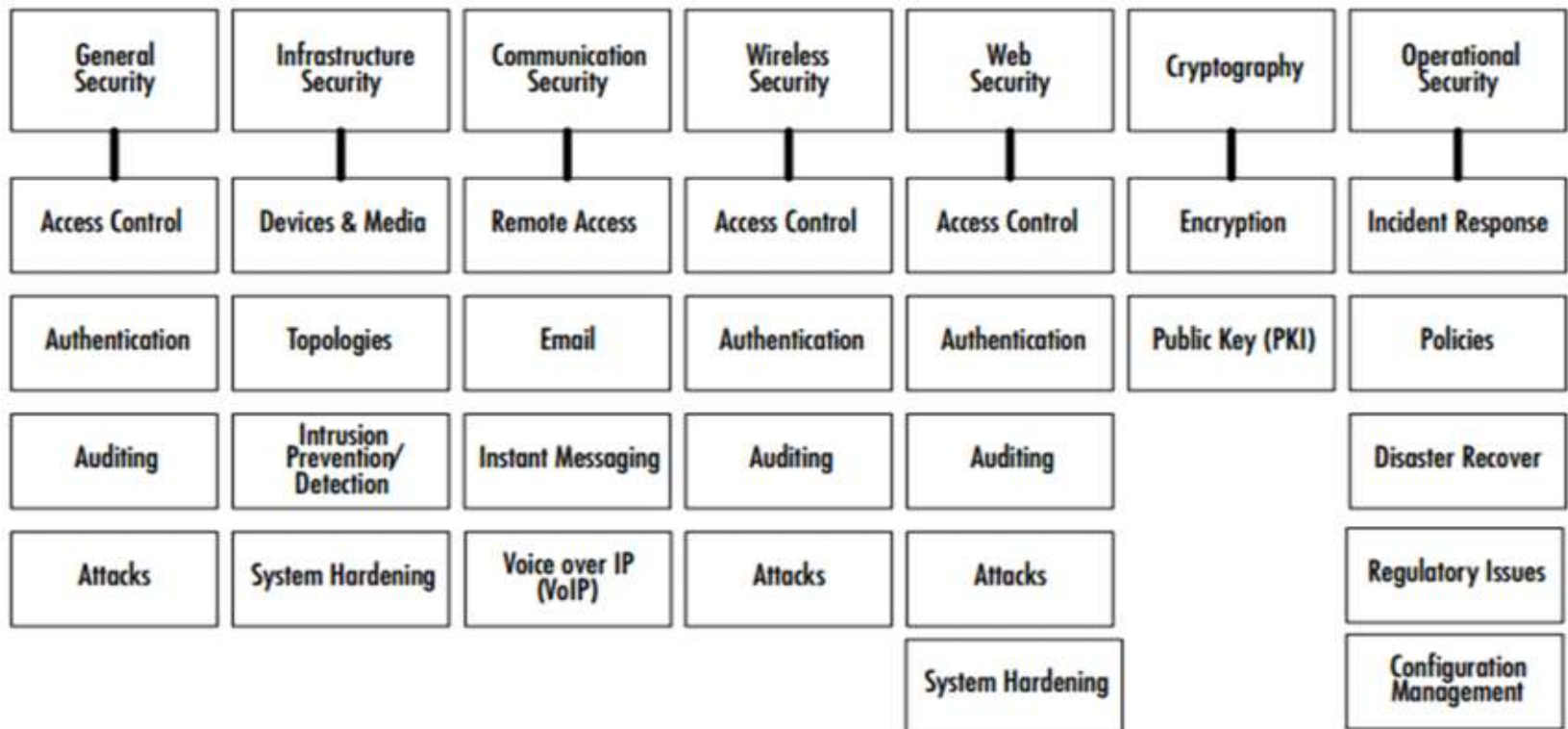
Maintain Network Security

7

Protect your network from unauthorised access. Use firewalls against external threats and filter out malicious content. Test your security's robustness.

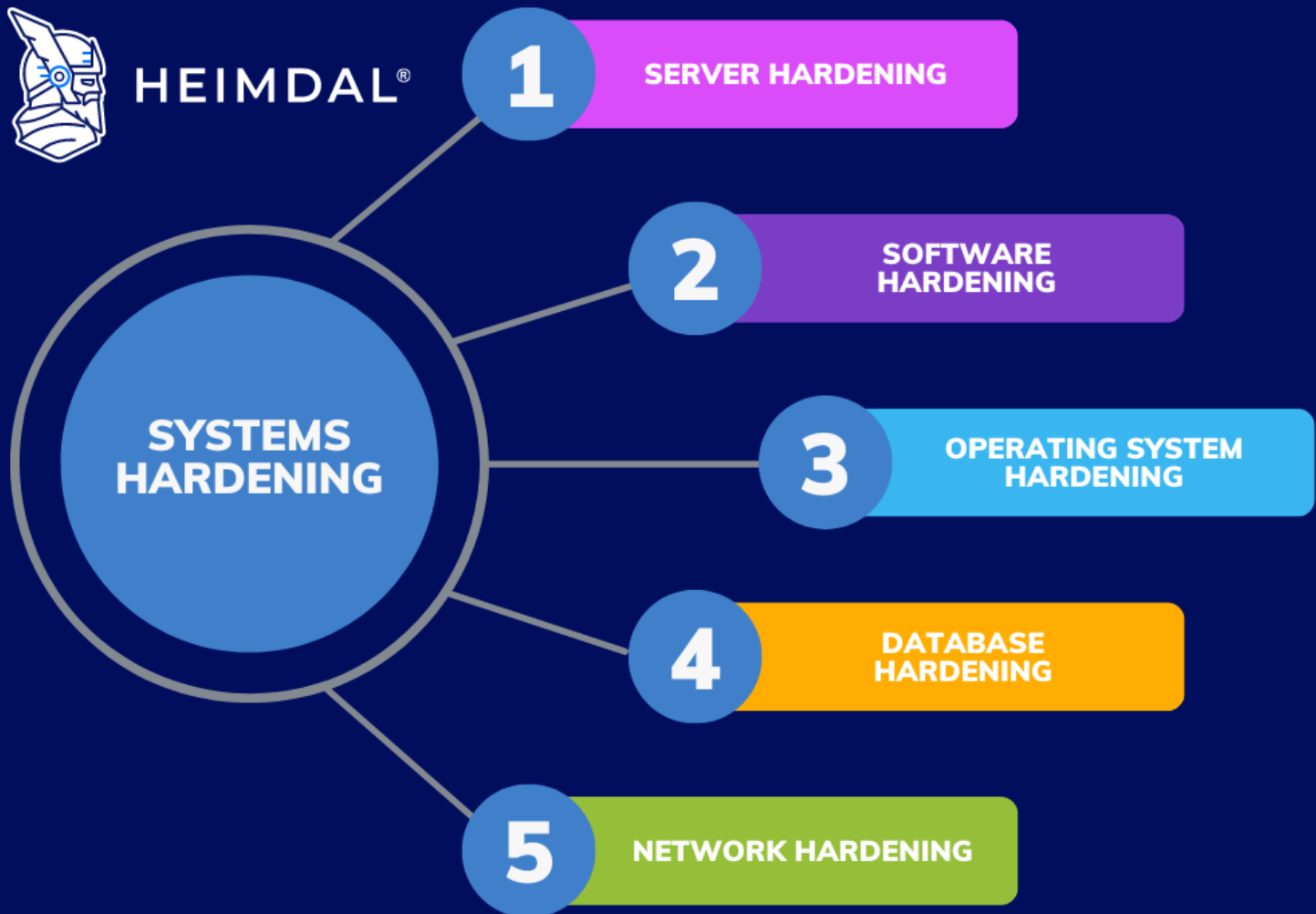


Information security categories:



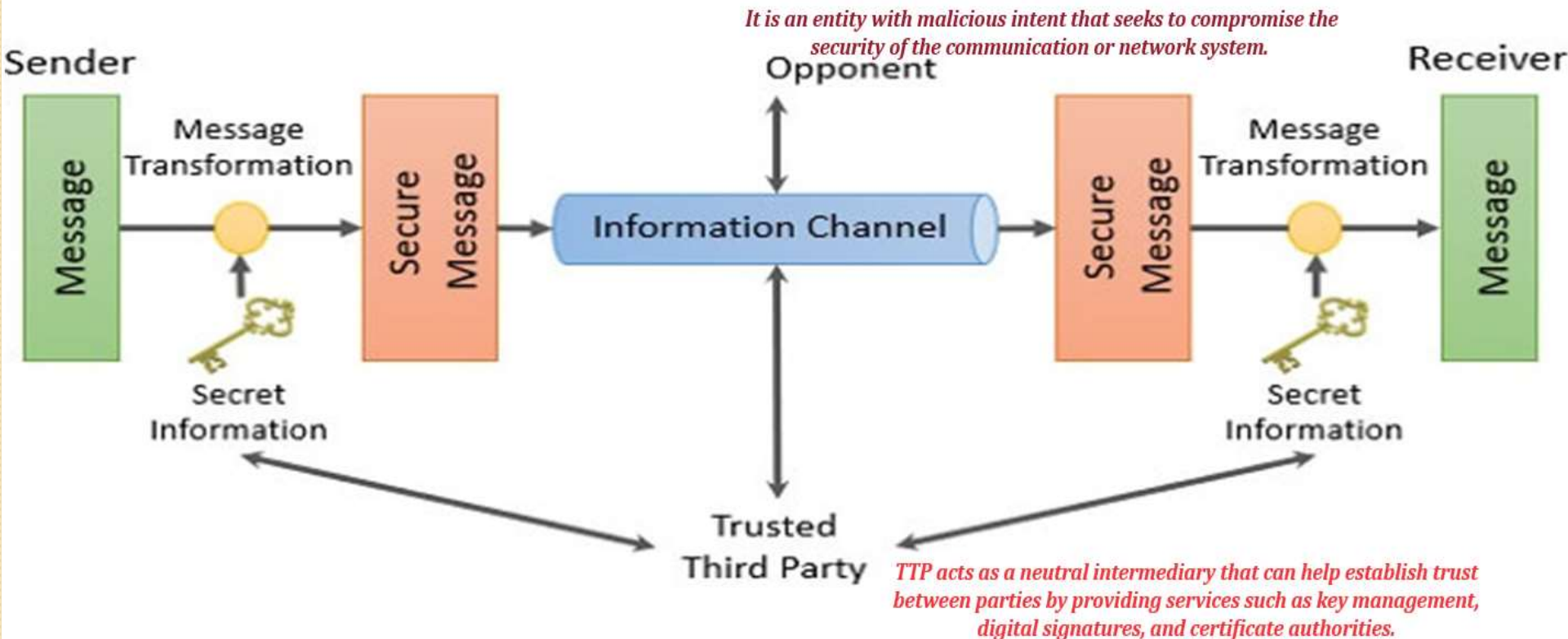


HEIMDAL®



Operational Model for Network Security

- A **Network Security Model** exhibits how the **security service** has been designed over the network
- to prevent the opponent from causing a threat to the confidentiality or authenticity of the information transmitted through the network.



Model for Network Security

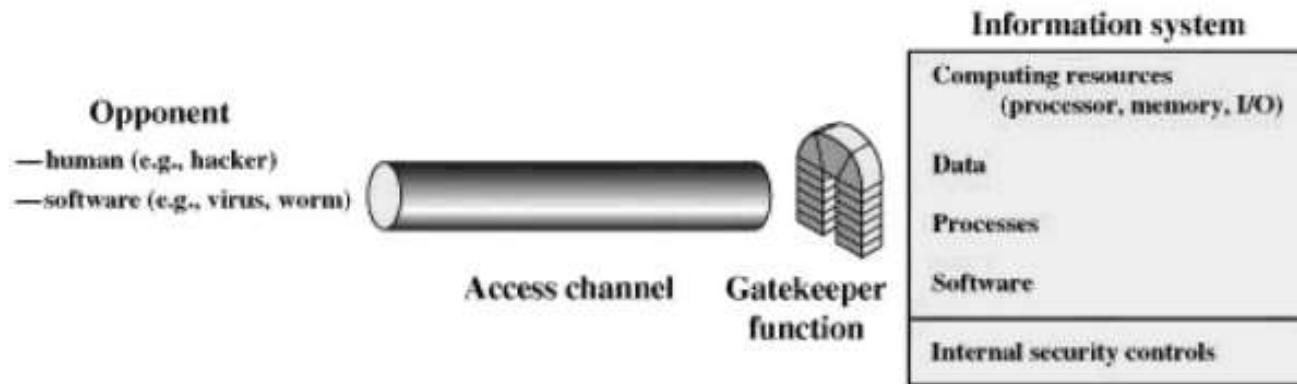
Model Requirement:

1. **Algorithm** for the security transformation
2. generate the secret information (**keys**) used by the algorithm
3. develop **methods to distribute** and share the secret information
4. specify a **protocol** enabling the principals to use the transformation and secret information for a security service

- A message is to be **transferred from one party to another** using Internet service.
- The two parties sender and receiver, they must **cooperate for the exchange** to take place.
- A logical information channel is established by **defining a route through the Internet from source to destination** and by the cooperative use of communication protocols (e.g., TCP/IP)
- Original message of the sender going to **transform a readable message at the sender side into an unreadable format**, an appropriate algorithm should be designed such that it should be difficult for an opponent to crack that security algorithm.
 - **Secret information** is used while transforming the message which will also be required when the message will be retransformed at the recipient side.
 - That's why a trusted third party is required which would take the responsibility of distributing this secret information to both the parties involved in communication
- An appropriate methodology must be used to deliver the secret information to the communicating parties **without the interference of the opponent** (use of communication protocols (e.g., TCP/IP)
- When receiver received an message then **cipher text converted to plain text using decryption algorithm**

Model for Network Access Security

- To protect **the transmission of information from a potential threat** to its confidentiality, integrity, and more.



- This is important to secure **sensitive data** safe from cyber attacks and to ensure that the network is usable and can be trusted
- Network security protects the data of the client by ensuring that hackers can not get into your network easily.

Cont..

- You are well aware of the attackers who attack your system that is accessible through the internet. **These attackers fall into two categories:**
- **Hacker:** The one who is only interested in penetrating into your system. They do not cause any harm to your system they only **get satisfied by getting access to your system.**
- **Intruders:** These attackers **intend to do damage to your system** or try to obtain the information from the system which can be used to attain financial gain.

Cont..

- The attacker can place a **logical program** on your system through the network which can affect the software on your system. This leads to two kinds of risks:
 - **a. Information threat:** This kind of threats **modifies data** on the user's behalf to which actually user should not access. Like enabling some crucial permission in the system.
 - **b. Service threat:** This kind of threat **disables** the user from accessing data on the system.

Cont..

- There are two ways to secure your system from attacker of which the first is to introduce the **gatekeeper function**. Introducing gatekeeper function means introducing **login - id** and **passwords** which would keep away the **unwanted access**.
- In case the unwanted user gets access to the system the second way to secure your system is introducing **internal control** which would detect the unwanted user trying to access the system by analyzing system activities.
- This second method we call as **antivirus** which we install on our system to prevent the unwanted user from accessing your computer system through the internet.



Model for Network Access Security

Model Requirements:

1. Select appropriate **Gatekeeper functions** to identify users.
2. Implement **security controls** to ensure only authorized users access designated information or resources.

Threat and Vulnerabili



- First important to learn what is an **asset** in the world of cyber security.
- **Asset: Asset includes people, property, and information.**
 - People: **employees** and other **stakeholders** of an organization
 - Property: both tangible and intangible items carrying some value
 - Information: databases, software code, company accounts and records, etc.
- **Every now and then, these assets are exposed to a threat, risk, and vulnerability.**

What is Vulnerability?

- Any weakness in an information system, system processes, or internal controls of an organization.
- The attackers can use this weakness to gain access to your systems.
- It act as a gateway or entry point for future cyber-attacks and intrusions.
- Hackers will constantly test out these areas of weakness and can exploit them if they see an opportunity.
- Examples of vulnerability:
 - A weakness in a firewall
 - Lack of security cameras
 - Unlocked doors at businesses

Examples



1. **Outdated software:** **not up to date** can compromise the security of the host system and leave the entire network vulnerable.
Example: **outdated antivirus software or operating system.**
2. **Misconfigured Systems:** If a system is not configured according to the **organisation's security policy**, that would also result in Vulnerabilities. **Example:** Using a system that **doesn't contain the organisation's safety and monitoring tools.**
3. **Unprotected APIs:** APIs act like a bridge between **applications and the internet.**
4. **Example:** Sharing access to your cloud drives with unauthorized individuals or trying to access them.

Cont..

4. **Weak Credentials:** same set of passwords for almost all applications and logins.
Example: 1234, Abcde or like using Birthdays and names like James1956.
4. **Lack of Strong Encryption:** Although encryption can't prevent Hacking, it can slow down the attacking momentum.
5. **Example:** Try encrypting files using a password when transferring data. This option is available in simple file types like WinZip and WinRAR.
6. **Zero Day Vulnerabilities:** A zero-day Vulnerability is a flaw or set of weaknesses uncovered by an attacker but is unknown to the software developer and the organisation that uses the software.
7. **Example:** Any attack that went unnoticed from a few days to several weeks until it's discovered can be considered a Zero-day Vulnerability.
8. **Unauthorized Access: Organization** gives its employees more access than necessary, it could also lead to a data breach.
9. **Example:** Restricting access to install, modify or remove any software tool

Threats

- Threat is any potential hazard that is likely to harm an organization, whether intentionally or unintentionally, associated with the exploitation of an existing vulnerability.
- Three categories:
 - **Intentional threats** are actions or methods that are used by bad actors to compromise a security or software system.
 - Examples: malware, malicious code.
 - **Unintentional threats** are often caused by human error.
 - For example, an employee might forget to lock the door to the IT servers or leave sensitive information unmonitored.
 - **Natural threats** are unpredictable events such as floods, earthquakes, and other natural disasters that have the potential to damage an organization's assets.

Risk

- Risk is the probability of loss of asset, exposure to threat, and potential damage from a cyber attack.
- It is basically the meeting point of threat and vulnerability.



Examples of risk include:

1. Financial losses
2. Loss of privacy
3. Legal implications
4. Even loss of life

Risk can also be defined as:

Risk = Asset + Threat x Vulnerability



Key concepts

- **Basic Terminologies
in Network Security**
- **Threats and
Vulnerability**

Key Concepts

- Subject vs Object
- Asset
- Risk
- Control, Safeguard, or Countermeasure
- Vulnerability
- Threat
- Threat Agent
- Attack
- Exploit
- Exposure
- Loss
- Protection profile or Security posture

Subject vs Object

- **Subjects:** Active entity that require to access to an **Object**, A computer can be both the subject and object of an attack, for example, it is compromised by an attack (object), and is then used to attack other systems (subject).



Subject



objects



Subject



objects



Asset

- The organizational resource that is being protected. An asset can be logical, such as a Web site, information, or data; or an asset can be physical, such as a person, computer system, or other tangible object.



website



Data



computers



workers



building

Risk

- The probability that something unwanted will happen. Organizations must minimize risk to match their **risk appetite**—the quantity and nature of risk the organization is willing to accept.



Earthquake



Office without Security Guards



Thunderstorm

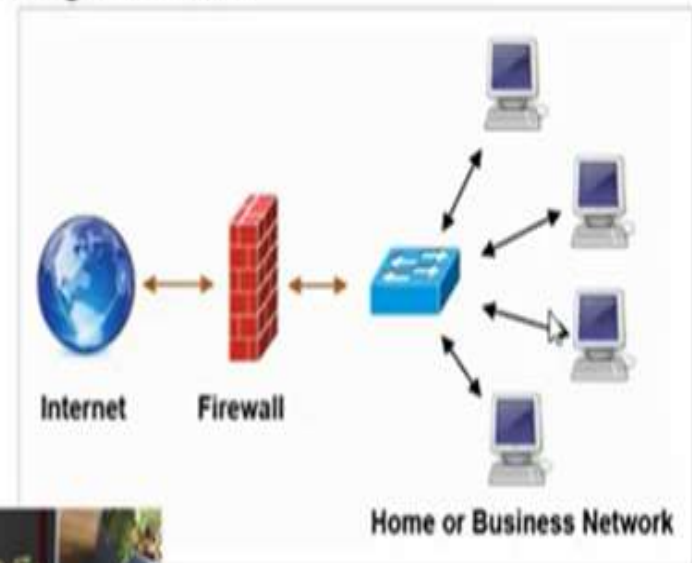


Flood



Control, Safeguard, or Countermeasure

- Security mechanisms, policies, or procedures that can successfully counter attacks, reduce risk, resolve vulnerabilities, and otherwise improve the security within an organization.



Vulnerability

- A weaknesses or fault in a system or protection mechanism that opens it to attack or damage.



Computer without updated patches



Threat and Threat Agent

- A category of objects, persons, or other entities that presents a danger to an asset. Threats are always present and can be purposeful or undirected.
- **Threat agent:** The specific instance or a component of a threat.



Computer without updated patches

Attack

- An intentional or unintentional act that can cause damage to or otherwise compromise information and/or the systems that support it. Attacks can be active or passive, intentional or unintentional, and direct or indirect.

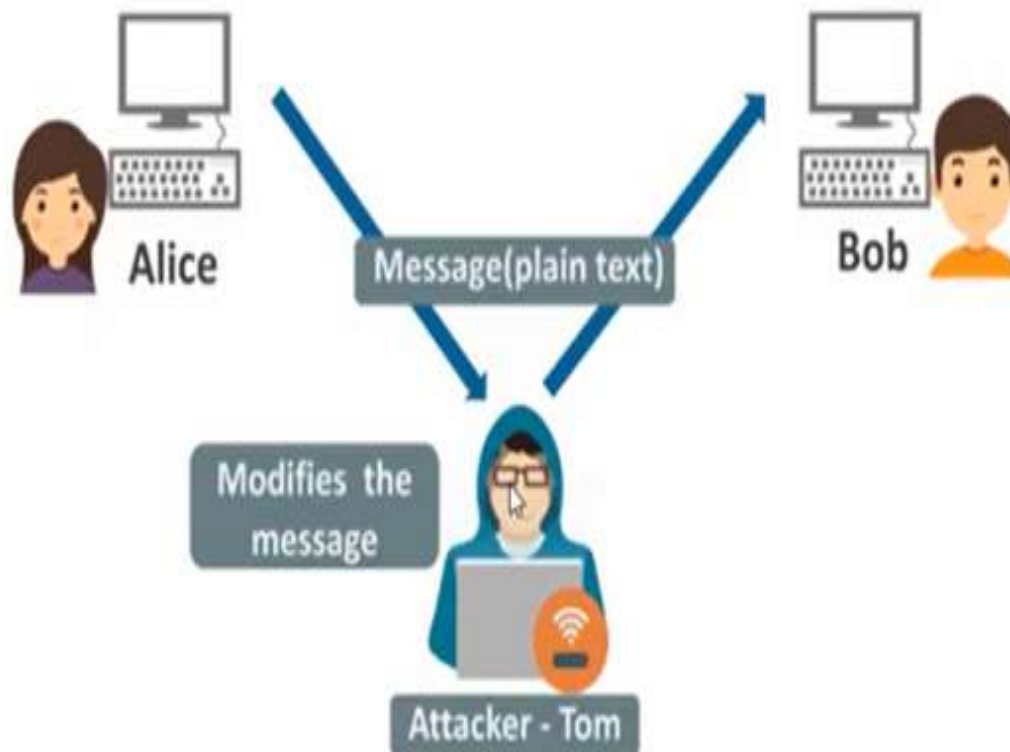


Computer without updated patches

Attack

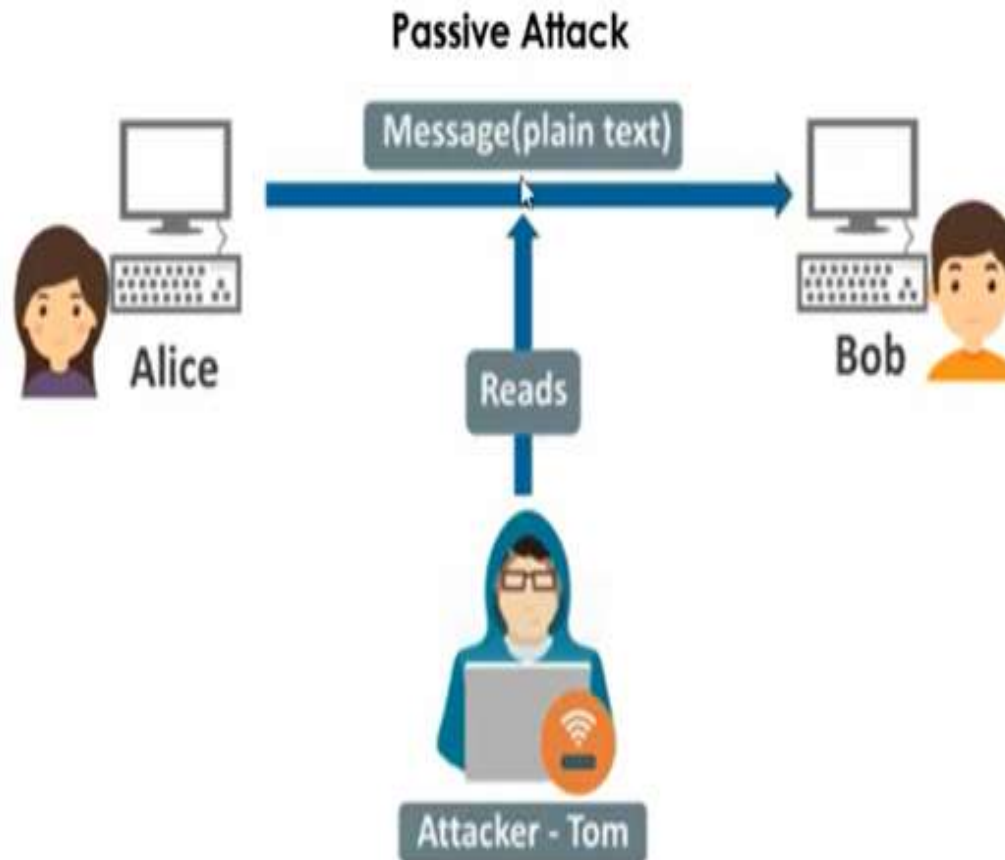
- An intentional or unintentional act that can cause damage to or otherwise compromise information and/or the systems that support it. Attacks can be active or passive, intentional or unintentional, and direct or indirect.

Active Attack



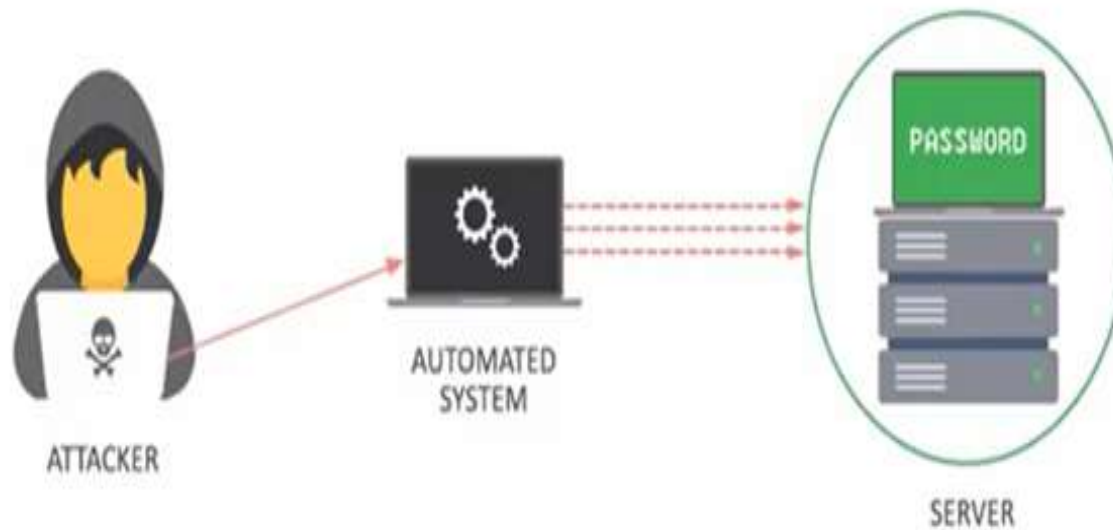
Attack

- An intentional or unintentional act that can cause damage to or otherwise compromise information and/or the systems that support it. Attacks can be active or passive, intentional or unintentional, and direct or indirect.



Exploit

- A technique used to compromise a system. Threat agents may attempt to exploit a system or other information asset by using it illegally for their personal gain.



Exposure

- A condition or state of being exposed. In information security, exposure exists when a vulnerability known to an attacker is present.



Loss

- A single instance of an information asset suffering damage or unintended or unauthorized modification or disclosure. When an organization's information is stolen, it has suffered a loss.



Protection profile or security posture

- The entire set of controls and safeguards, including policy, education, training and awareness, and technology, that the organization implements (or fails to implement) to protect the asset.



What is Security?

- Security is defined as State of **being free from danger or threat.**
- Freedom from **threat.**
- **In general**, Security means being free from danger.
- To be secure is to be protected from the **risk of loss, damage, unwanted modification or other hazards.**



Security is Everywhere

- Computer Security
- Home Security
- Health Security
- Human Security
- Food Security
- National Security
- Corporate Security
- Cloud Security
- Ecological Security

Guess the Picture



What is Privacy

- **The state** ,in which one is not being observed or disturbed by other.



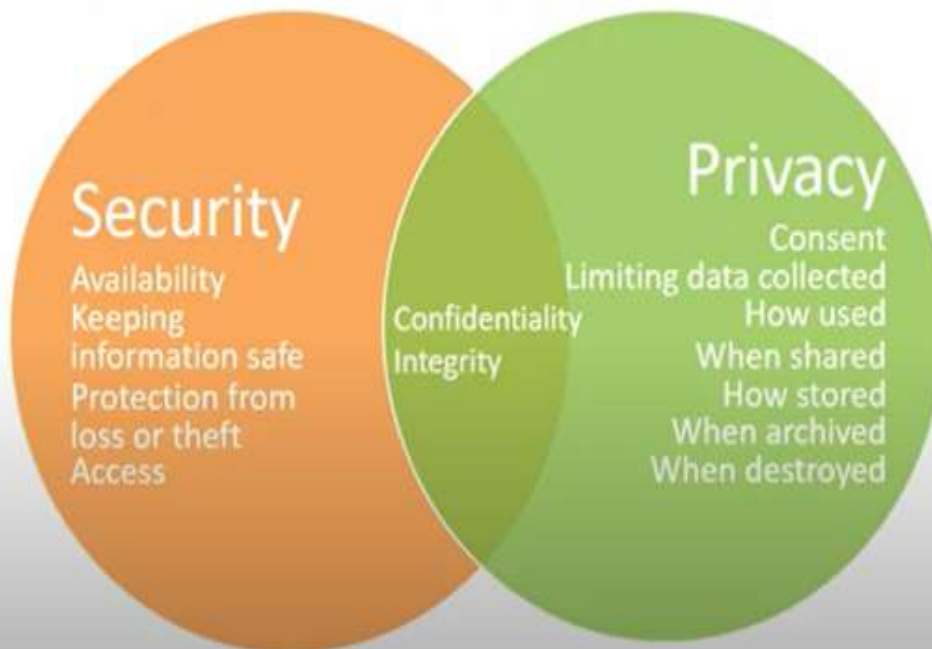
Security and Privacy

What is Privacy

A state in which one is not observed or disturbed by other

Security is the technical methods used to protect the data and not concerned with how and when it used.

Privacy is how an organization processes Personal Data to comply with laws, regulations.



- Security can be achieved without privacy but privacy can not be achieved without Security.



Security VS Privacy

Security

Security refers to protection against unauthorized access.

Security Provides protection for all types of data and information including the ones that are stored electronically.

Security program focuses on all sorts of information assets an organization collects.

It implements security protocols to provide confidentiality, integrity and availability of information assets

Security can be achieved without privacy.

Privacy

Privacy defines the ability to protect personally identifiable information

Privacy means protecting sensitive information related to individuals and organizations

Privacy program focuses on personal information such as names, addresses, login credentials, financial accounts, etc.

It refers to protection of privacy rights with respect of processing of personal data.

Privacy cannot be achieved without security.

VS



Extra Part.....

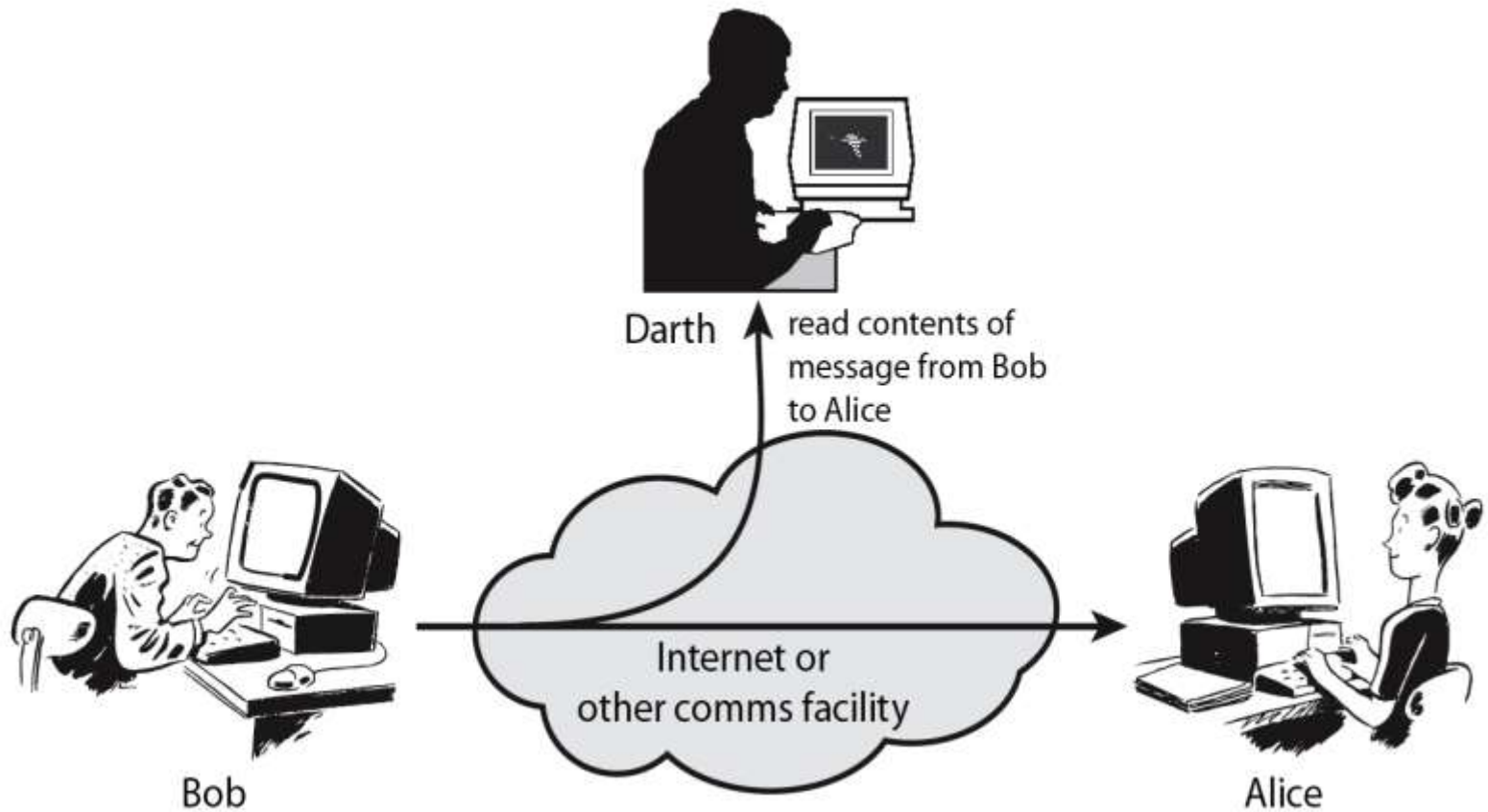
Aspects of Security

- consider **3 aspects** of information security:
 - **security attack**
 - **security mechanism/techniques**
 - **security service**

Security Attack

- any action that compromises the security of information owned by an organization
- generic types of attacks
 - Passive
 - Active

Passive Attacks





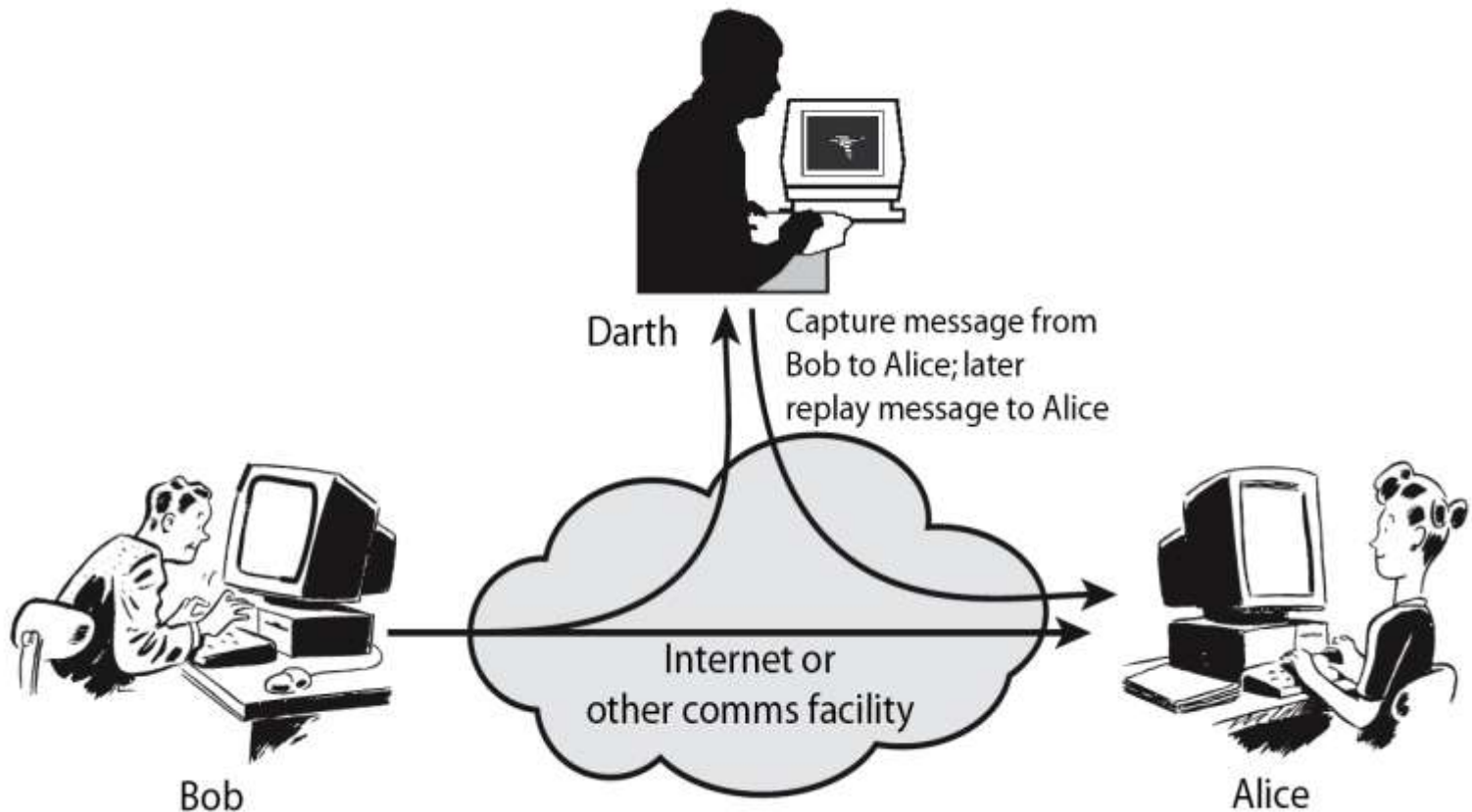
“passive attacks” attempt to learn or make use of information from the system but does not affect **system resources**.

By eavesdropping on, or monitoring of, transmissions to:

- + obtain message contents, or**
- + monitor traffic flows**

Are difficult **to detect** because they **do not involve any alteration** of the data.

Active Attacks



- 
- “**active attacks**” attempt to **alter system resources or affect their operation.**
 - By modification of data stream to:
 - + **masquerade** of one entity as some other
 - + **replay** previous messages
 - + **modify** messages in transit
 - + **denial of service(DoS)**
 - It is quite difficult **to prevent active attacks** absolutely, because of the wide variety of potential physical, software, and network vulnerabilities.

Security Attacks: Active and Passive attacks in Information Security

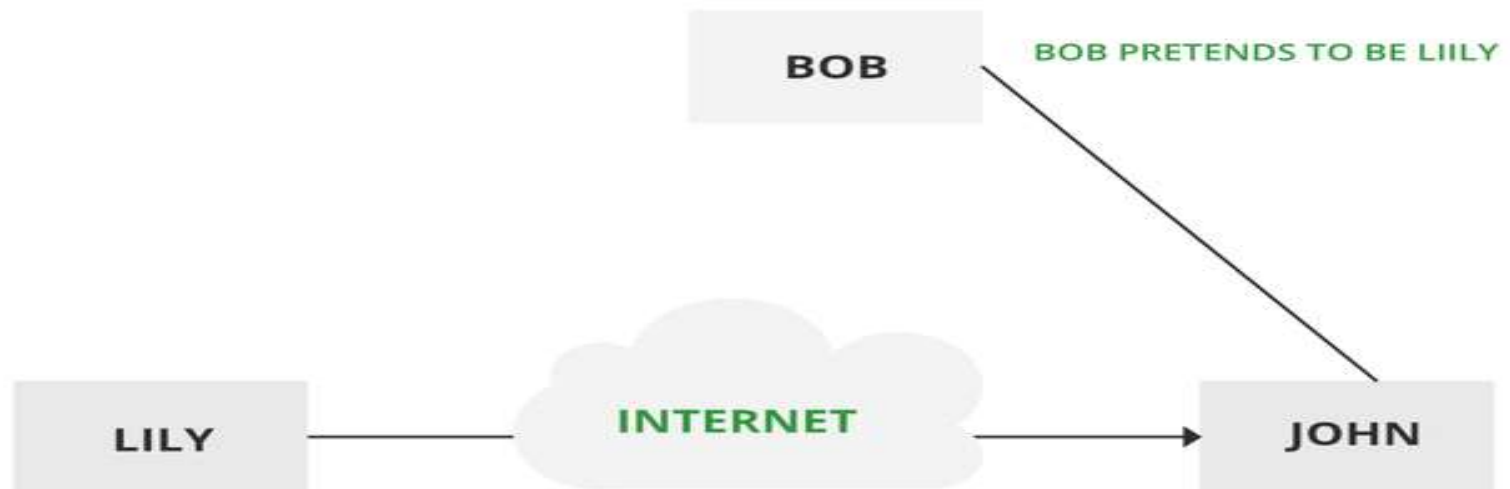
Active attacks: An Active attack attempts to alter system resources or affect their operations.

Active attacks involve some modification of the data stream or the creation of false statements.

- Types
- Masquerade
- Modification of messages
- Repudiation
- Replay
- Denial of Service

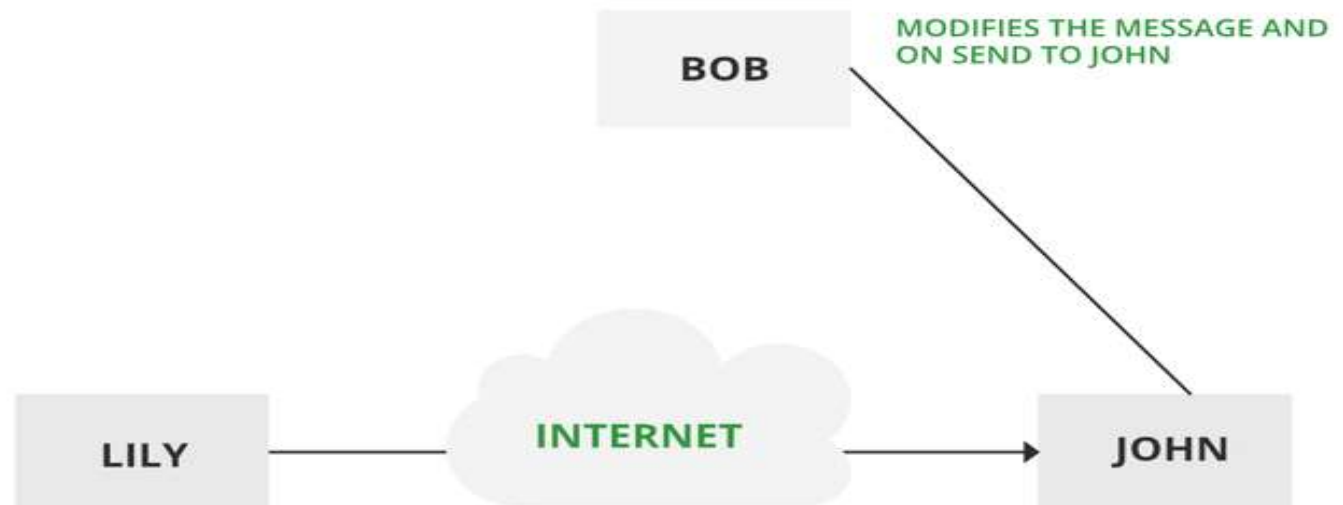
Masquerade –

- A masquerade attack takes place when **one entity pretends to be a different entity.**
- If an authorization procedure isn't protected, it is liable to a masquerade assault.
- Ex: **stolen passwords and logins**
- **finding gaps in programs**
- **locating a manner across the authentication process.**



Modification of messages –

- It means that **some portion of a message is altered** or that message is delayed or reordered to produce an unauthorized effect.
- Modification is an attack on the **integrity** of the original data.
- It means that unauthorized parties not only gain access to data but also spoof the data by triggering denial-of-service attacks, such as **altering transmitted data packets or flooding the network with fake data.**

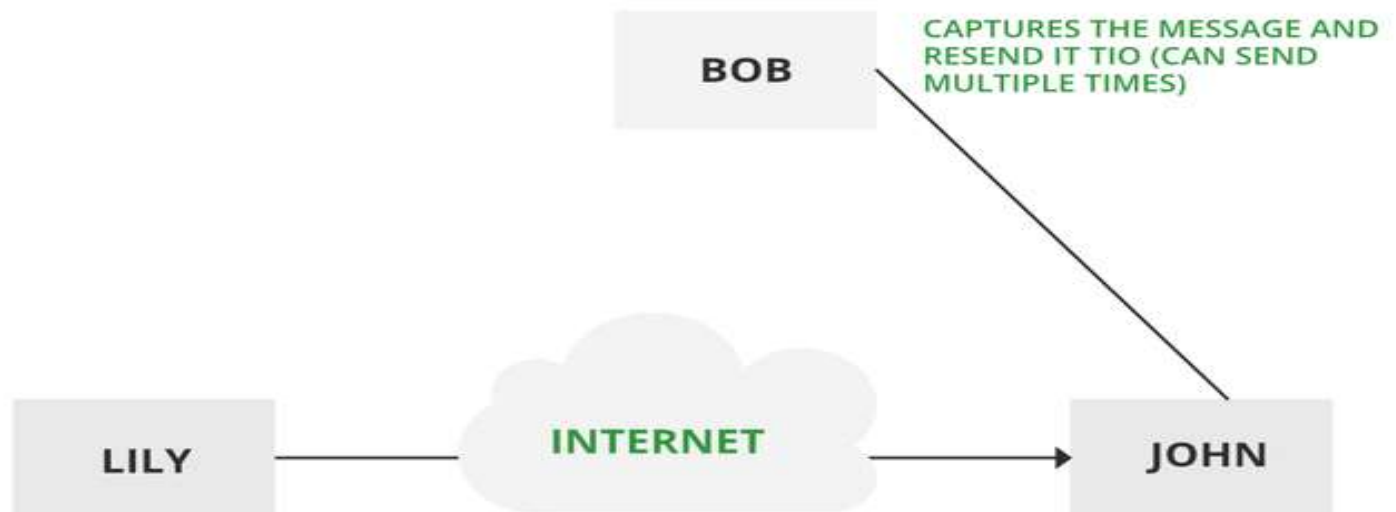


Repudiation –

- This attack occurs when the network is **not completely secured or the login control** has been tampered with.
- With this attack, the author's information can be **changed by actions of a malicious user** in order
- **Ex: to save false data in log files, up to the general manipulation ,**
- **spoofing of e-mail messages.**

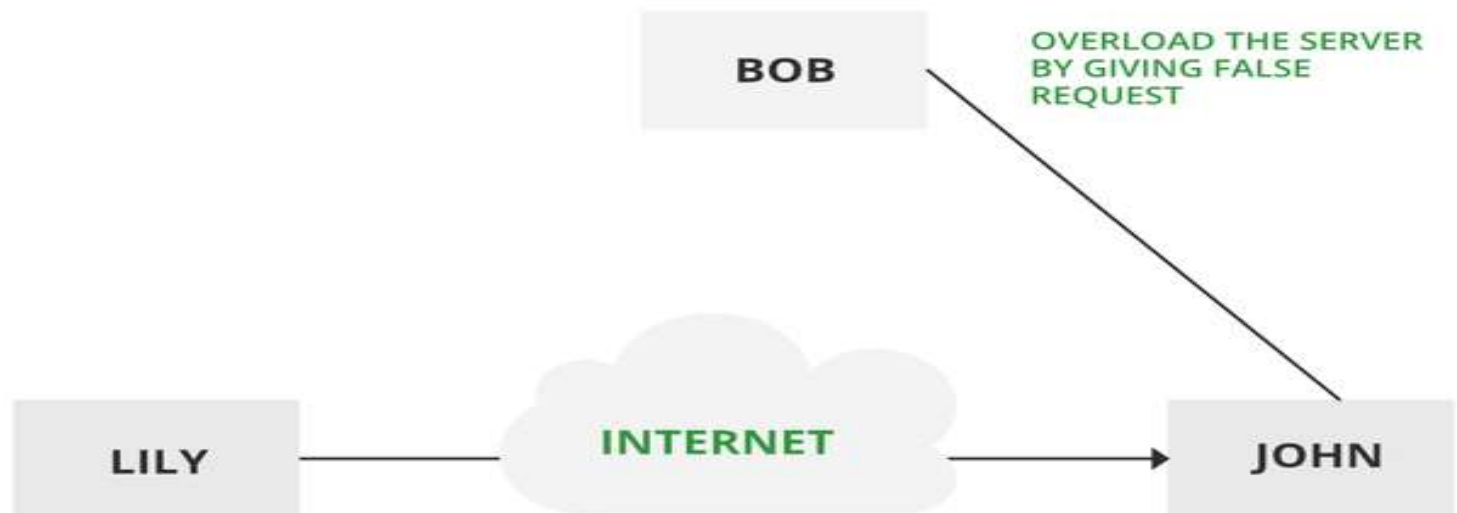
Replay –

- It involves the passive capture of a message and its subsequent transmission to produce an authorized effect.
- Aim: to save a copy of the data originally present on that particular network and later on use this data for personal uses.
- Once the data is corrupted or leaked it is insecure and unsafe for the users.



Denial of Service –

- It prevents the normal use of communication facilities. This attack may have a specific target.
- For example, an entity may suppress all messages directed to a particular destination.
- Another form of service denial is the disruption of an entire network either by disabling the network or by overloading it with messages so as to degrade performance.

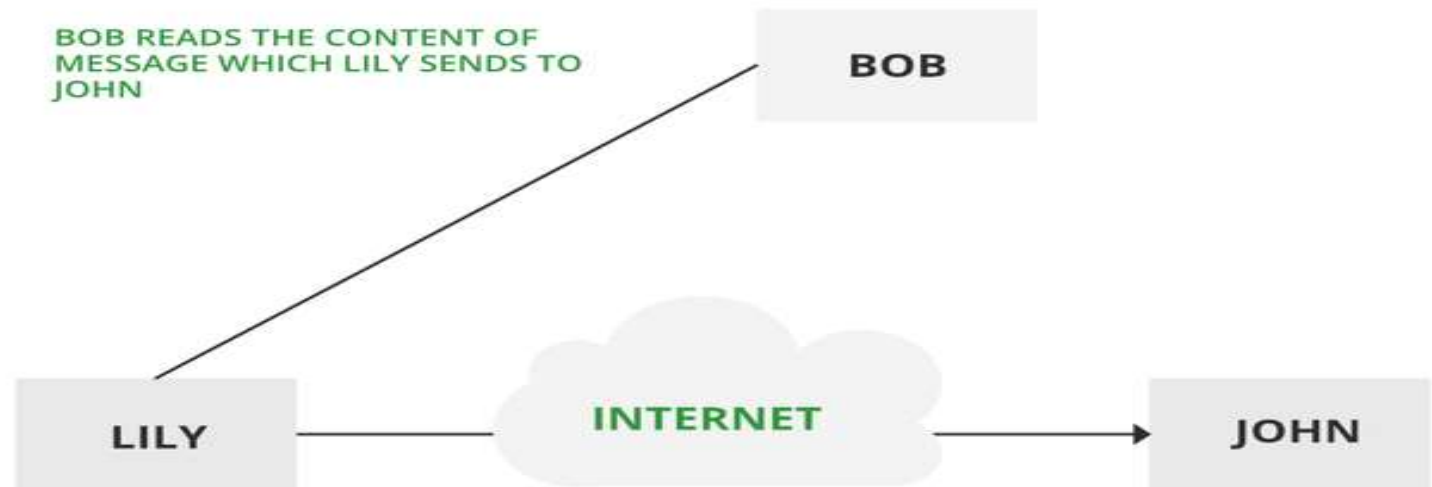


Passive attacks:

- A Passive attack attempts to learn or make use of information from the system but does not affect system resources.
- Passive Attacks are in the nature of eavesdropping on or monitoring transmission.
- The goal of the opponent is to obtain information that is being transmitted.
- Types
 - The release of message content
 - Traffic analysis

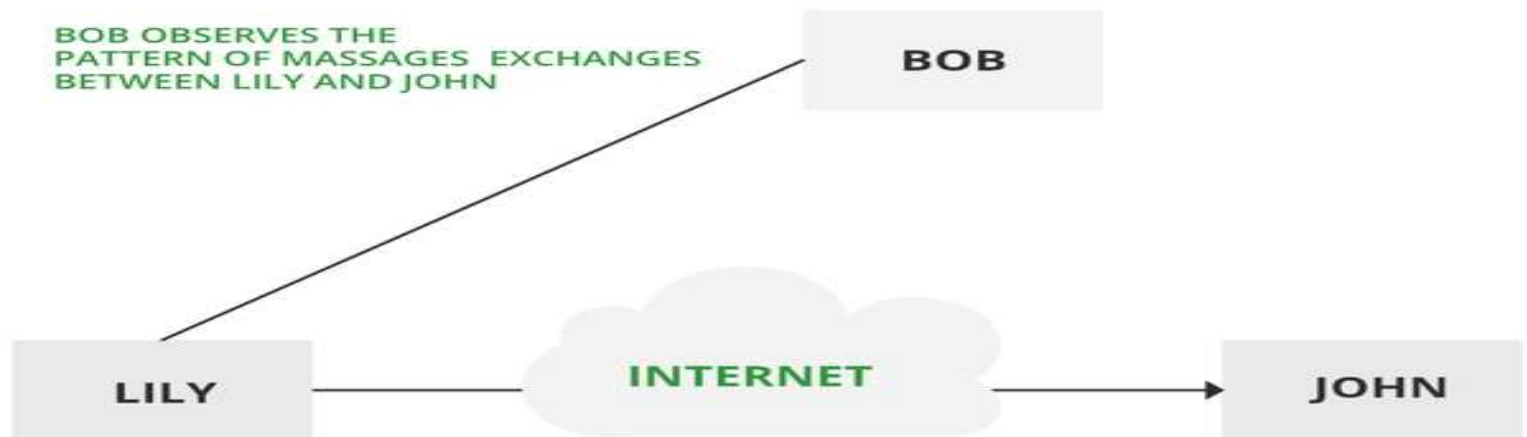
The release of message content –

- Telephonic conversation, an electronic mail message, or a transferred file may contain **sensitive or confidential information.**
- We would like to prevent an opponent from **learning the contents of these transmissions.**



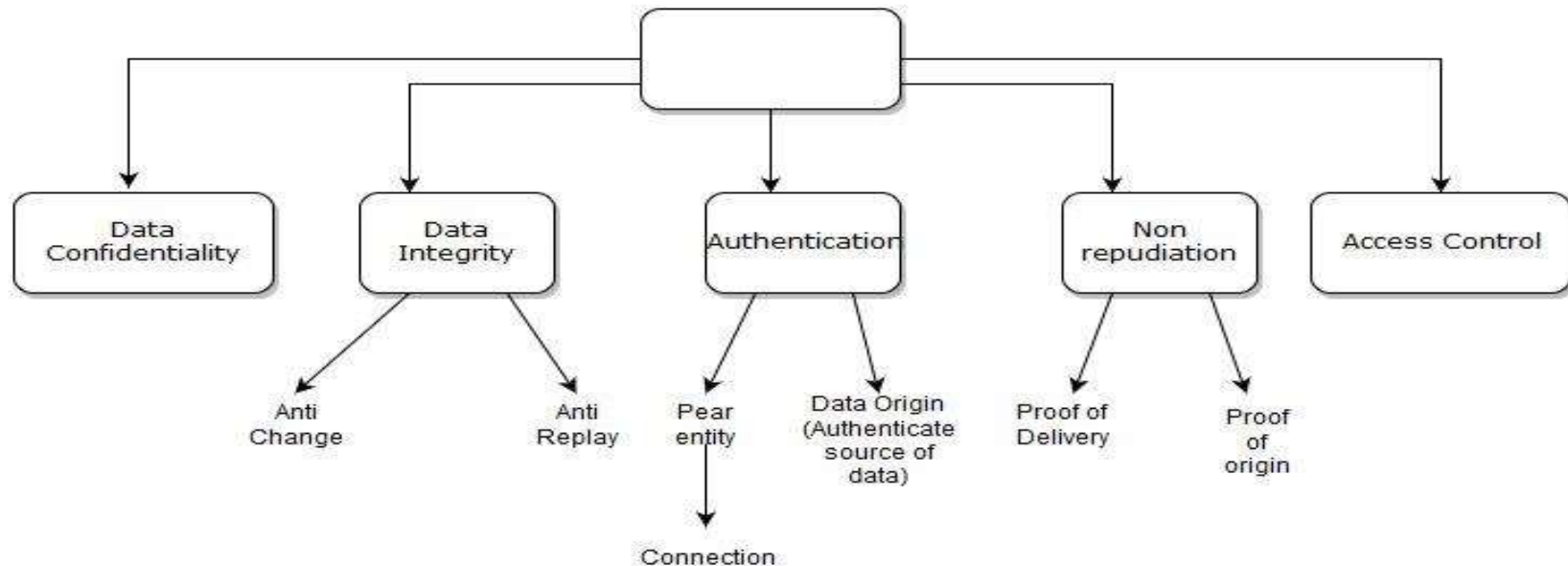
Traffic analysis –

- Suppose we use encryption technique, in such case also...
- The opponent could determine the **location and identity** of **communicating host** and could observe the **frequency and length of messages** being exchanged.
- This information might be **useful in guessing the nature of communication** that taking place.



Security Services:

- that enhances the security of the data processing systems and the information transfers of an organization.
- These services make use of one or more security mechanisms to provide the service.
- Following are the five categories of these services:



- **Data Confidentiality:** Protects data from **unauthorized disclosure**.
- **Data Integrity:** The assurance that **data received are exactly as sent by an authorized entity** (i.e., contain no modification, insertion, deletion, or replay).
- **Authentication:** The **assurance** that the **communicating entity** is the one that it claims to be.
- **Peer Entity Authentication:** **identity of the entities connected**.
- **Data-Origin Authentication:** In a connectionless transfer, provides assurance that the **source of received data is as claimed**.
- **Non-repudiation:** Protects against **denial by one** of the entities involved in a communication of having participated in all or part of the communication.
- **Proof of Origin:** Proof that the message was sent by **the specified party**.
- **Proof of Delivery:** Proof that the message was **received by the specified party**.
- **Access Control:** The prevention of **unauthorized use of a resource** (i.e., this service controls who can have access to a resource, under what conditions access can occur, and what those accessing the resource are allowed to do).



THANK YOU

Assignment no. I

Q1) a) List and explain various elements of Information Security. [5]

b) Write difference between security and privacy. [5]

OR

Q2) a) Enlist the security goals and mechanisms. [5]

b) Draw and explain operational security model for network security. [5]

Q3) a) Define cyber security. Explain Active and passive attack. [5]

Unit-1 Introduction

Class: TE -C

Question Bank

Q.No	Questions
1	What are the elements of Information Security? Explain in Detail
2	What is CIA Triad? Explain with diagram
3	Explain the following Terms: 1) Threat 2) Vulnerability
4	Elaborate different security Techniques
5	Differentiate between Active Attack and Passive Attack
6	What is security service? Explain different security services in detail
7	Define Information System. Explain components of Information System
8	Differentiate between Security and Privacy
9	Explain operational model of network security with help of Diagram
10	Explain the types of Security mechanisms